increased number of data sources and teams, the central governance team becomes a bottleneck. Furthermore, too many complex problems requiring the attention of a group often lead to the loss of the bigger picture. Finally, a centralized model, requiring formal communication, leads to overgrown bureaucratic overhead on even the simplest operations and modifications needed at the local level, aggressively slowing development.

DECENTRALIZED DATA GOVERNANCE MODEL: BOTTOM-UP DECISION FLOW

We can implement decentralized governance in two ways. One is leaving data ownership to software system product owners, and the other is regulating data governance with all kinds of data committees, guilds, or councils.

Such empowerment of product owners has some benefits. For one, each system can be quickly changed and adapted. Also, such a setup often leads to open development platforms, where anyone can experiment and provide business with new tools, products, and services. Furthermore, decentralized structures are believed to be more meritocratic. Last but not least, the independence of systems leaves the whole setup less likely to contain a single point of failure.

However, this garden of roses does not come without a thorn or three. First, each system team working independently on its infrastructure protocols may cause major disconnections, and the big picture of owned data may be lost. Second, maintaining the expertise required to fulfill legal obligations, for example, becomes multiplied or missing, at the same time increasing the cost and lowering chances of covering the entirety of data operations. Third, data infrastructure initiatives lack the power of the scale, and differences in technology stacks and formats lead to an increased difficulty of cross-domain analysis.

6.4.3 Federated data governance model

As this chapter title suggests, a federated data governance model is the one that the data mesh proposes as a way to combine the strengths of two extreme models and eliminate their weaknesses. If this model is well executed, its main benefit results from combined strengths of the central